Evaluating Expressions with Radicals

Question Bank - Additional 4.05

Name_____
Date_____
Period_____

Show your work. Circle your final answer.

For questions 1 through 6, evaluate.

1. $\sqrt{64}$

2. $\sqrt[3]{8}$

3. $\sqrt[3]{-125}$

4. $\sqrt{\frac{9}{49}}$

5. $\sqrt{4^2+33}$

6. $\sqrt[3]{5^2+100}$

For questions 7 and 8, evaluate. Each expression has a power and a radical.

7. Evaluate each expression.

a. $\sqrt{144} \cdot 3^{-2}$

b. $\left(-\frac{1}{4}\right)^2 + \sqrt{3^2 + 27}$

c. $\sqrt[3]{-27} - 2^{-1}$

8. Evaluate each expression.

a. $\sqrt[3]{64} - 1.5\left(\sqrt{7^2 - 13}\right)$

b. $\left(-\frac{1}{2}\right)^3 - \sqrt[3]{4^2 + 11}$

c. $\sqrt{6^2 + 13} \cdot 2^{-3}$

Working backwards.

9. Find the value of n that makes each statement true. (The radicand is the unknown here, so it is not one of the listed perfect values until you find it.)

a. $\sqrt{n} = 15$

b. $\sqrt[3]{n} = -2$

c. $\sqrt{n-5} = 11$

Reading someone else's work.

10. Deshawn writes $\sqrt{100-36} = \sqrt{100} - \sqrt{36} = 4$. Evaluate the expression correctly, and explain in one sentence what the radical bar does that his method ignores.

11. Which is the value of $\sqrt{8^2+36}$?

- ○ A. 14
- ○ B. 10
- ○ C. 100
- ○ D. 50

12. Select ALL of the expressions whose value is 4.

- ☐ A. $\sqrt{2^3+17}$
- ☐ B. $\sqrt[3]{64}$
- ☐ C. $\sqrt[3]{-64}$
- ☐ D. $\sqrt{16}$
- ☐ E. $\sqrt{5^2-9}$
- ☐ F. $\sqrt[3]{3^3+37}$